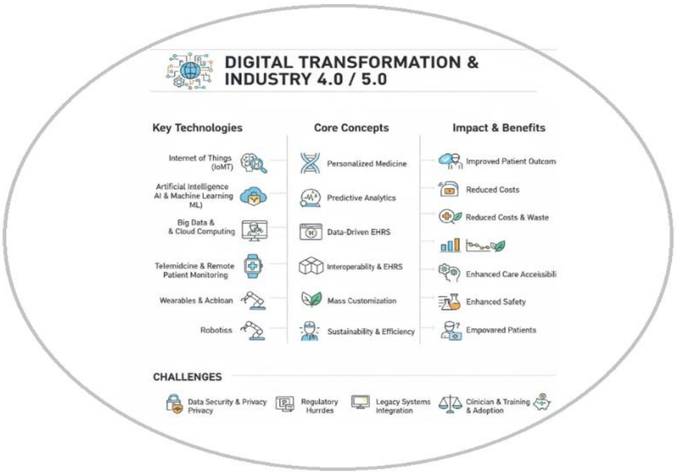
# Balochistan University of Engg & Tech Khuzdar

**BIOMEDICAL ENGINEERING DEPARTMENT**




**DIGITAL TRANSFORMATION & INDUSTRY 4.0/5.0 TECHNOLOGIES**

**LAB- BME-362 (6th SEMESTER)**

**Name:** ________________________________________

**Roll.NO:** ________________________________________

**Submitted to:** **Dr. Wazir Muhammad**

# BALOCHISTAN UNIVERISTY OF ENGINEERING&TECHNOLOGY, KHUZDAR

**Biomedical Engineering Department**

| **BUET Vision** |
|---|
| To become a world class higher education Institute leading to socio economic development of the region and beyond |
| **BUET Mission** |
| To impart quality education and research with professional excellence on strong ethical foundation toserve the region and beyond |
| **Biomedical Engineering Department Vision** |
| 1. To be a leading biomedical engineering department that produces advanced technology and research for human well-being.<br>2. To be a renowned center for scientific research that improves the quality of human life.<br>3. To be the premier biomedical engineering platform in the world, based on excellence in people, research, and education. |
| **Biomedical Engineering Department Mission** |
| 1. To equip graduates with diverse skills to create enabling technologies and new knowledge for improving human health.<br>2. To provide a stimulating educational environment that enables students to develop engineering solutions for the healthcare sector.<br>3. To train students in the integration of science, engineering, and medicine to create clinically translatable solutions for human health. |

# BALOCHISTAN UNIVERISTY OF ENGINEERING &TECHNOLOGY, KHUZDAR

**Biomedical Engineering Department**

| Program Learning Outcomes | |
|---|---|
| PLO1 | **Engineering Knowledge**: An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems. |
| PLO2 | **Problem Analysis:** An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. |
| PLO3 | **Design/Development of Solutions**: An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. |
| PLO4 | **Investigation**: An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions. |
| PLO5 | **Modern Tool Usage**: An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations. |
| PLO6 | **The Engineer and Society:** An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems. |
| PLO7 | **Environment and Sustainability**: An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development. |
| PLO8 | **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice |
| PLO9 | **Individual and Team Work**: An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings. |
| PLO10 | **Communication**: An ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. |
| PLO11 | **Project Management**: An ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment. |
| PLO12 | **Lifelong Learning**: An ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments. |

# Engineering Lab Rubric

Outcomes Assessed

1. Conduct of Experiment 2. Ethics 3. Individual and Teamwork
4. Conduction of experiment (Software)

| Sr. No | | Rubrics | Skills Level | | |
|---|---|---|---|---|---|
| | | | **Exceeds Expectation** | **Meets Expectation** | **Doesn't Meets Expectation** |
| | | | **2.5** | **1.25** | **0.83** |
| **1** | **Conduction of experiment (Hardware)** $R_1 = 2.5$ | **Selection of Equipment = 0.833** | Student has selected correct equipment relevant to the experiment = 0.833 | Student needed minor guidance to select correct equipment relevant to the experiment. = 0.416 | Student is incapable of selectin equipment relevant to the experiment. = 0.277 |
| **2** | | **Equipment Connection = 0.833** | Student has made correct equipment/component connections as per standard circuit diagrams. | Student needed guidance to make correct equipment/component connections as per standard circuit diagram | Student was unable to make correct equipment/component connections as per standard circuit diagrams. |
| **3** | | **Results = 0.833** | Accurate results have been achieved. | The achieved results are not accurate but are within tolerance range | No results are achieved OR the achieved results are meaningless. |
| **4** | **Ethics** $R_2 = 2.5$ | **Safety** | Student carefully observes the safety rules and procedures during practical work | Student observes safety rules and procedure with minor deviation during practical work | Student does not care about safety rules during practical work. |
| **5** | **Team Work** $R_3 = 2.5$ | **Listen to other teammates** | Consistently listens and responds to others appropriately | Usually doing most of the talking, rarely allowed others to speak | Student shows an assertive behavior and was unable to show respect towards other teammates. |
| **6** | **Conduction of Experiment (Software)** $R_4 = 2.5$ | **Familiarity with software** | Student has full command on the basic tools of the software | Student has limited command on the basic tools of the software | Student has no idea how to use the basic tools of the software |

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(Autonomous)

**Biomedical Engineering Department**

## *Certificate*

*This is to Certify that it is a bonafide record of Practical work done by Mr/Mis. ___________________________________ Bearing the Roll No. _____________________ of ____________ Class _____________________________________ Branch in the ___________________________ laboratory during the Academic year ________________ Under our supervision.*

**Head of the Department** **Lab-Incharge**

# BALOCHISTAN UNIVERISTY OF ENGINEERING & TECHNOLOGY, KHUZDAR

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**Biomedical Engineering Department**

## Assessment Record

**Session:**________________ **Lab Venue:** **Computer Lab**

| Course Title | Roll No. | Student Name | Department |
|---|---|---|---|
| Digital Transformation & Industry 4.0/5.0 Technologies | | | **BMED** |

| S.NO | Lab Objectives | CLO | PLO | Domain Level | Marks | Signature |
|---|---|---|---|---|---|---|
| 1 | To **evaluate** the impact of AI medical robots (Nurabot, GR3, Moxy, Pepper) on reducing nurse workload and improving patient interaction through MATLAB simulation. | 4 | 4 | 4 | | |
| 2 | To **observe** and analyze the operational mechanisms of key robotic applications, including robotic-assisted surgery, rehabilitation robotics, telepresence, pharmacy automation, and robotic prosthetics. | 3 | 3 | 3 | | |
| 3 | **Design** Continuous Stress Detection in Nurses Using Wearable Sensors with Machine Learning Approach | 5 | 5 | 5 | | |
| 4 | To perform the Remote Patient Monitoring (RPM) System Demonstration and Operation. | 3 | 3 | 3 | | |
| 5 | To observe the Implementation of an IoT-Based Child Monitoring System with Real-Time Video Streaming. | 3 | 3 | 3 | | |
| 6 | To implement a simple encryption and decryption algorithm for data security purpose using Python. | 3 | 3 | 3 | | |
| 7 | **Design** a code for Diabetes Prediction Using Deep Learning with TensorFlow and Keras | 5 | 5 | 5 | | |
| 8 | **Design** and Implementation of Python Tensor Flow Keras model to classify the Brain Tumor using Convolutional Neural Networks. | 5 | 5 | 5 | | |
| 9 | To **develop** and train a CNN model in MATLAB for accurate classification of breast cancer images into benign and malignant categories. | 5 | 5 | 5 | | |
| 10 | To simulate a heartbeat monitoring system using Arduino and a heartbeat sensor in Proteus. | 3 | 3 | 3 | | |
| 11 | To Design and evaluation of a Virtual Reality Brain Surgery Simulator for Training and User Experience Assessment. | 6 | 6 | 6 | | |
| 12 | To design an electronic medical health record for a patient using Healthbridge simulation software. | 4 | 4 | 4 | | |
| 13 | To perform the setting up of 3D Printing Lab | 4 | 4 | 4 | | |
| 14 | To understand the basic structure and functioning of a blockchain and implement a simple blockchain with multiple blocks using Python. | 3 | 3 | 3 | | |